

Experiment No-1

Student Name: HARSH

Branch: MCA

Semester: 3rd

Subject Name: Business Analytics Lab

UID: 24MCA20045

Section/Group: 24MCA-1(B)

Date of Performance: 25/07/2025

Subject Code: 24CAH-703

1. Aim of the practical:

Formatting the worksheet using logical formula IF Statement, Nested IF, AND, OR, NOT, IFERROR, SUMIF, AVERAGEIF, COUNTIF and AVERAGEIF). To understand and implement various logical and conditional functions in Microsoft Excel for analyzing student data. The key goal is to use formulas such as IF, Nested IF, AND, OR, NOT, IFERROR, SUMIF, AVERAGEIF, and COUNTIF to evaluate conditions, handle errors, and perform conditional calculations for effective decision-making.

2. Overview:

In this practical, a worksheet containing student details (Name, Marks, Attendance) is formatted using various logical and conditional formulas. The objective is to derive meaningful insights such as pass/fail status, grading, attendance issues, and performance summaries. Here's how each formula is applied:

3. Apparatus:

- **Microsoft Excel (2016 or above) :**
Primary tool for creating and formatting the worksheet using formulas.
- **Excel Dataset (Student Marks Sheet) :**
Predefined student data (Name, Marks, Attendance) to work on.

4. Procedure:

Given Columns:

NAME, MARKS, ATTENDANCE

a. Pass/Fail (IF):

Formula (in cell D2):

=IF(B2>=40,"Pass","Fail")

If marks are 40 or above, return "Pass"; else, "Fail"

b. Grade (Nested IF):

Formula (in cell E2):

=IF(B2>=80,"A",IF(B2>=60,"B",IF(B2>=40,"C","Fail")))

Assign grade based on marks:

≥80 → A

≥60 → B

≥40 → C

<40 → Fail

c. Pass Both (AND):

Formula (in cell F2):

=IF(AND(B2>=40, C2>=50),"Pass", "Fail")

Pass only if both marks ≥ 40 and attendance ≥ 50.

d. Pass Any (OR):

Formula (in cell G2):

=IF(OR(B2>=40, C2>=50),"Pass", "Fail")

Pass if either marks or attendance conditions are met.

e. Low Attendance (NOT):

Formula (in cell H2):

=NOT(C2>=50)

Returns TRUE for low attendance (i.e., <50%), FALSE otherwise.

f. Safe Division (IFERROR):

Formula (in cell I2):

=IFERROR(B2/C2,"Error")

Prevents divide-by-zero errors by showing "Error" if denominator is 0.

g. Total > 50 (SUMIF):

Formula (in cell J2):

=SUMIF (B2:B12,">50", B2:B12)

Adds only the marks that are > 50.

h. Average > 50 (AVERAGEIF):

Formula (in cell K2):

=AVERAGEIF (B2:B12,">50", B2:B12)

Finds average marks only for students scoring > 50.

i. Count > 50 (COUNTIF):

Formula (in cell L2):

=COUNTIF (B2:B12,">50")

Counts how many students scored more than 50 marks.

5. Output:

	A	B	C	D	E	F	G	H	I	J	K	L
1	NAME	MARKS	ATTENDANCE	PASS/FAIL(IF)	GRADE(NESTED IF)	Pass Both(AND)	Pass Any(OR)	Low Attendance(NOT)	Safe Division(IFERROR)	Total>50(SUMIF)	Average>50(AVERAGEIF)	Count>50(COUNTIF)
2	HARSH	90	80	PASS	A	PASS	PASS	FALSE	1.125	457	77.42857143	7
3	RAHUL	92	65	PASS	A	PASS	PASS	FALSE	1.415384615	452	75.33333333	6
4	VARUN	79	70	PASS	B	PASS	PASS	FALSE	1.128571429	360	72	5
5	SAURAV	30	90	FAIL	FAIL	FAIL	PASS	FALSE	0.333333333	281	70.25	4
6	PREETY	80	40	PASS	A	FAIL	PASS	TRUE	2	281	70.25	4
7	YASH	60	75	PASS	B	PASS	PASS	FALSE	0.8	201	67	3
8	DINESH	50	89	PASS	C	PASS	PASS	FALSE	0.561797753	141	70.5	2
9	SAHIL	25	56	FAIL	FAIL	FAIL	PASS	FALSE	0.446428571	141	70.5	2
10	UMESH	56	46	PASS	C	FAIL	PASS	TRUE	1.217391304	141	70.5	2
11	ANANYA	85	24	PASS	A	FAIL	PASS	TRUE	3.541666667	85	77.42857143	1

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6. Learning outcomes (What I have learnt):

- Gain hands-on experience with logical operators and conditional calculations.
- Learn to handle errors like divide-by-zero using IFERROR.

- Understand how to summarize data using conditional aggregation (SUMIF, AVERAGEIF, COUNTIF).
- Improve spreadsheet skills for real-world data analysis in education or business.

Evaluation Grid:

Sr. No.	Parameters	Marks Obtained	Maximum Marks
1.	Demonstration and Performance		12
2.	Worksheet		8
3.	Via		10